SOFTWARE REQUIREMENTS SPECIFICATION

for

AstraLib

AI-Powered Universal Knowledge Discovery and Library Management Platform

Version 2.0

Prepared by

K. Sarika
K. Chaitra
L. Gayathri
M. Poojitha
M. Manogna

Keshav Memorial Institute of Technology (KMIT)
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Revision History

Name	Date	Reason For Changes	Version
K. Sarika, K. Chaitra, L. Gayathri, M. Poojitha, M. Manogna	2026	Initial SRS development	1.0
K. Sarika, K. Chaitra, L. Gayathri, M. Poojitha, M. Manogna	2026	Updated architecture, modules, AI capabilities and requirements	2.0

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of AstraLib, an AI-Powered Universal Knowledge Discovery and Library Management Platform.

AstraLib is designed to modernize conventional digital library systems by combining traditional library management capabilities with Artificial Intelligence (AI), Retrieval-Augmented Generation (RAG), semantic search, intelligent document understanding, personalized recommendations, document summarization, citation support, and collaborative learning.

The system is intended to serve a broad range of users including learners, students, researchers, engineers, doctors, professionals, educators, and other literate users from different disciplines. The platform is not restricted to a single educational stream or user category.

The SRS defines the system's:

- Functional requirements
- Non-functional requirements
- User roles
- System features
- External interfaces
- AI capabilities
- Security requirements
- Data requirements
- Performance expectations
- Business rules
- Assumptions and dependencies

This document serves as a reference for developers, testers, project guides, administrators, librarians, and other stakeholders during system design, implementation, testing, deployment, and maintenance.

1.2 Document Conventions

The following conventions are used throughout this document.

Requirement identifiers

Functional requirements use FR-XX, where XX represents the requirement number. Non-functional requirements use NFR-XX, where XX represents the requirement number.

Requirement terminology

- Shall – Mandatory requirement.
- Should – Recommended requirement.
- May – Optional capability.
- TBD – To Be Determined.

Priority

System features are classified as:

- High – Essential for the core system.
- Medium – Important supporting functionality.
- Low – Future or optional functionality.

Technical terminology

Technical terms such as AI, RAG, LLM, embedding, vector database, semantic search, and API are defined in Appendix A.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders.

Developers

Developers can use this document to understand system functionality, architecture, modules, APIs, data requirements, AI functionality, and security requirements.

Project Guide / Supervisor

The project guide can use the SRS to evaluate project scope, the problem being addressed, the proposed solution, system requirements, technical feasibility, and expected functionality.

Testers

Testers can use the functional and non-functional requirements to design test cases and verify system behavior.

Librarians

Librarians can understand digital resource management, content verification, categorization, moderation, reports, and analytics.

Users

Users can understand the services available through AstraLib, including search, reading, AI assistance, recommendations, Study Hall, and document interaction.

Administrators

Administrators can understand user management, role management, system configuration, AI configuration, security, analytics, backup, and recovery.

Suggested Reading Order

Readers are recommended to follow this order: (1) Introduction, (2) Overall Description, (3) External Interface Requirements, (4) System Features, (5) Nonfunctional Requirements, (6) Other Requirements, (7) Appendices.

1.4 Product Scope

AstraLib is a web-based AI-powered library and knowledge discovery platform designed to transform conventional library systems into intelligent knowledge ecosystems.

Traditional library management systems primarily provide cataloguing, circulation, and keyword-based retrieval. As digital resources increase, users face difficulties in discovering relevant resources, understanding lengthy documents, and obtaining contextual answers.

AstraLib addresses these limitations by providing:

- Traditional digital library management
- Keyword-based search
- Semantic search
- Natural-language search
- AI conversational assistance
- Retrieval-Augmented Generation
- Document question answering
- AI-generated summaries
- Personalized recommendations
- Reading history
- Favorites
- Citation and evidence support
- Digital document management
- Study Hall collaboration
- Real-time discussion
- Notifications
- Analytics
- Administrative management
- Content moderation
- Secure role-based access

Major Objectives

- Improve knowledge discovery.
- Reduce the time required to locate relevant resources.
- Provide contextual natural-language interaction with digital resources.
- Enable users to ask questions directly from documents.
- Provide personalized resource recommendations.
- Improve utilization of digital books, journals, papers, and technical documents.
- Support collaborative learning through Digital Study Hall.
- Reduce repetitive work for librarians.
- Provide intelligent analytics for administrators.
- Preserve essential library management functionality while introducing AI capabilities.

Expected Benefits

- Faster information discovery.

- Better search relevance.
- Improved document understanding.
- Personalized knowledge discovery.
- Better utilization of digital resources.
- Reduced manual effort.
- Improved collaboration.
- Better decision-making through analytics.
- Accessible knowledge across multiple disciplines.

1.5 References

The following references are relevant to the development of AstraLib:

- IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998.
- ISO/IEC/IEEE 29148 – Systems and Software Engineering – Life Cycle Processes – Requirements Engineering.
- Karl E. Wiegers, Software Requirements Specification Template.
- “Transforming OPACs into Intelligent Discovery Systems: An AI-Powered, Knowledge Graph-Driven Smart OPAC for Digital Libraries.”
- Research literature related to Artificial Intelligence in Digital Libraries.
- Research literature related to Semantic Search and Information Retrieval.
- Research literature related to Retrieval-Augmented Generation.
- Research literature related to Knowledge Graphs.
- React.js documentation.
- Spring Boot documentation.
- Python and FastAPI documentation.
- MySQL documentation.
- ChromaDB documentation.
- Gemini API documentation.
- Research literature related to recommender systems and personalized information retrieval.

2. Overall Description

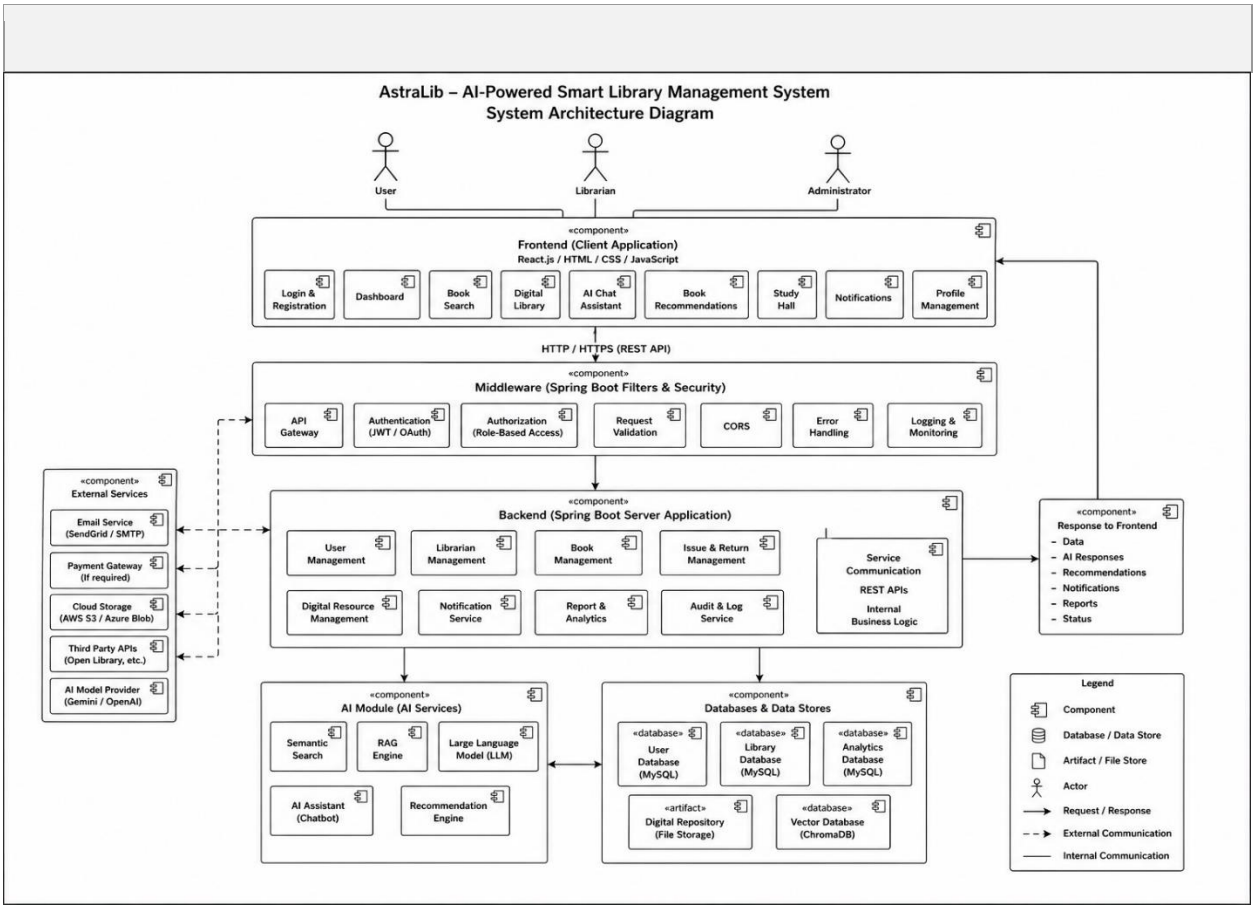
2.1 Product Perspective

AstraLib is a new-generation digital library and knowledge discovery platform.

Traditional OPAC and library systems primarily depend on keyword matching and structured metadata. The base paper motivating AstraLib highlights the limitations of conventional search and proposes intelligent discovery through semantic retrieval and knowledge graphs.

AstraLib extends this concept by combining intelligent search with RAG, document understanding, recommendations, AI assistance, and collaborative learning.

The system follows a layered architecture, as shown below.



2.1.1 System Architecture Components

Frontend

The frontend provides the user interface.

Technology: React.js, HTML, CSS, JavaScript.

Major interfaces include:

- Login
- Registration
- Dashboard
- Search
- Digital Library
- AI Assistant
- Recommendations
- Study Hall
- Profile
- Favorites
- Reading History

Spring Boot Backend

The Spring Boot backend provides the core business logic. It manages:

- Authentication
- Authorization
- Users
- Books
- Categories
- Authors
- Digital resources
- Favorites
- Reading history
- Study Hall
- Notifications
- Reports
- Analytics

AI Service

A separate Python/FastAPI service manages AI functionality. It provides:

- Semantic search
- RAG
- Embedding generation
- AI chat
- Document question answering
- Summarization
- Recommendations
- Citation generation

MySQL

MySQL stores structured transactional information, examples include:

- Users
- Roles
- Books
- Authors
- Categories
- Favorites
- Reading history
- Study Hall
- Notifications
- Reports
- Audit logs

File Storage

File storage maintains:

- PDF books
- Research papers
- Journals
- Technical documents
- Notes
- Other supported digital resources

ChromaDB

ChromaDB acts as the vector database. It stores:

- Document embeddings
- Chunk representations
- Metadata
- Vector indexes

External AI Services

AstraLib may use external AI and knowledge services such as:

- Gemini API
- Open Library API
- OpenAlex API

2.2 Product Functions

The major functions of AstraLib are:

User Functions

- Registration
- Login
- Logout

- Password reset
- Profile management
- Interest management
- Keyword search
- Semantic search
- Resource filtering
- Book details
- Digital reading
- Favorites
- Reading history
- Reading progress
- AI chat
- RAG question answering
- Summarization
- Recommendations
- Citation viewing
- Study Hall
- Notifications

Librarian Functions

- Add resources
- Edit resources
- Delete resources
- Upload documents
- Manage authors
- Manage categories
- Manage metadata
- Verify uploaded content
- Moderate Study Hall
- View usage statistics
- Generate reports

Administrator Functions

- Manage users
- Manage librarians
- Manage roles
- Configure system settings
- Configure AI services
- Monitor security
- View analytics
- Manage backups
- Audit administrative operations

2.3 User Classes and Characteristics

2.3.1 General User

The general user represents any authorized individual using AstraLib for learning, research, professional development, or knowledge discovery. Examples include students, researchers, engineers, doctors, teachers, professionals, and lifelong learners.

Users should have basic knowledge of using a web browser. They can search resources, read digital content, ask AI questions, receive recommendations, join Study Hall, maintain favorites, and view reading history.

2.3.2 Librarian

The librarian manages the digital library collection. Responsibilities include resource management, metadata management, categorization, content verification, moderation, usage analysis, and report generation.

2.3.3 Administrator

The administrator has the highest privilege. Responsibilities include user management, librarian management, role assignment, system configuration, security, AI configuration, analytics, and backup and recovery.

2.4 Operating Environment

Client Environment

AstraLib operates through modern web browsers, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

Operating Systems

The application can be accessed through Windows, Linux, macOS, Android, and iOS, provided that a supported modern web browser is available.

Server Environment

Recommended server environment: Linux or Windows Server, minimum 4 GB RAM (8 GB or more recommended), multi-core processor, and sufficient storage for digital resources.

Software Environment

Component	Technology
Frontend	React.js
Backend	Spring Boot
AI Service	Python + FastAPI
Database	MySQL
Vector Database	ChromaDB
File Storage	Local/Cloud Object Storage

Component	Technology
AI Model	Gemini API / compatible LLM
API Communication	REST
Data Format	JSON
Authentication	JWT
Protocol	HTTPS

2.5 Design and Implementation Constraints

The following constraints apply:

- The system shall use a web-based architecture.
- Spring Boot shall be used for core backend services.
- React.js shall be used for the frontend.
- Python/FastAPI shall handle AI-specific processing.
- MySQL shall be used for structured data.
- ChromaDB shall be used for vector storage.
- Digital resources shall be stored separately from relational data.
- AI features depend on external or locally available AI models.
- Internet connectivity is required for cloud-based AI services.
- API keys shall not be exposed to frontend users.
- Uploaded documents must follow supported formats.
- User access shall be controlled through role-based authorization.
- AI responses must be grounded in retrieved information for document-based queries.
- The system must comply with applicable copyright and resource-access policies.
- The initial implementation shall prioritize core features required for the project timeline.

2.6 User Documentation

The following documentation shall accompany the system:

- User Manual
- Librarian Guide
- Administrator Guide
- Installation Guide
- Deployment Guide
- API Documentation
- AI/RAG Technical Guide
- Troubleshooting Guide
- Software Requirements Specification
- Software Design Document
- Testing Documentation

2.7 Assumptions and Dependencies

Assumptions

- Users have internet connectivity.
- Users have access to a modern browser.
- Librarians provide valid metadata.
- Uploaded documents are legally permitted for library use.
- Users have basic digital literacy.
- AI services are available when AI functionality is required.
- The database server is operational.
- Sufficient storage is available for digital resources.

Dependencies

AstraLib depends on:

- React.js
- Spring Boot
- Python
- FastAPI
- MySQL
- ChromaDB
- PDF processing libraries
- Embedding models
- Gemini or another supported LLM
- File storage
- Internet connectivity

3. External Interface Requirements

3.1 User Interfaces

The application shall provide responsive interfaces for:

Authentication

- Registration
- Login
- Logout
- Password reset
- Email verification

User Dashboard

The dashboard shall display:

- Recommended resources
- Recent resources
- Reading progress
- Favorites
- Study Hall activity
- Notifications

Search Interface

The search page shall support:

- Keyword search
- Natural-language search
- Semantic search
- Filters
- Sorting
- Resource categories

Digital Library

Users shall be able to:

- Browse resources
- View metadata
- Open documents
- Read PDFs
- Continue reading
- Bookmark resources

AI Assistant

The AI interface shall provide:

- Chat window
- Question input
- AI responses
- Source citations
- Document references
- Error/evidence messages

Study Hall

The interface shall support:

- Group creation
- Group joining
- Discussions
- Shared resources
- Notes
- Member activity

Librarian Dashboard

The librarian interface shall provide:

- Resource management
- Upload management
- Verification
- Moderation
- Reports

Administrator Dashboard

The administrator interface shall provide:

- User management
- Role management
- System settings
- AI settings
- Analytics
- Audit logs
- Backup controls

3.2 Hardware Interfaces

No specialized hardware is required. The system shall support desktop computers, laptops, tablets, and smartphones.

Basic peripherals: keyboard, mouse, touchscreen, and display. Optional hardware such as printers is not required for the core system.

3.3 Software Interfaces

Software	Purpose
React.js	User interface
Spring Boot	Backend services
FastAPI	AI service interface
MySQL	Structured data
ChromaDB	Vector search
Gemini API	LLM functionality
PDF Processing Library	Document extraction
Embedding Model	Vector generation
Open Library API	External book metadata
OpenAlex API	Scholarly metadata

3.4 Communications Interfaces

The system shall use HTTPS, REST APIs, JSON, and JWT authentication.

Communication between the following components shall be secured appropriately:

```
Frontend ↔ Backend
Backend ↔ AI Service
Backend ↔ Database
AI Service ↔ Vector Database
AI Service ↔ LLM API
```

Real-time Study Hall communication may use WebSocket, Server-Sent Events, or another suitable real-time communication mechanism.

4. System Features

4.1 User Authentication and Account Management

Description

This feature provides secure account creation and authentication.

Priority: High

Stimulus/Response

1. User opens registration.
2. User enters required details.
3. System validates the information.
4. System creates the account.
5. User logs in.
6. System authenticates credentials.
7. System generates an authenticated session/token.
8. User is redirected to the appropriate dashboard.

Functional Requirements

FR-01: The system shall allow new users to register using a valid email address and password.

FR-02: The system shall validate user information before account creation.

FR-03: The system shall authenticate registered users using secure credentials.

FR-04: The system shall maintain authenticated sessions securely.

FR-05: The system shall provide role-based access control.

FR-06: The system shall allow authenticated users to securely log out.

FR-07: The system shall allow users to initiate password recovery.

FR-08: The system shall provide secure password-reset functionality using a time-limited verification mechanism.

FR-09: The system shall support email verification where enabled.

4.2 User Profile and Personalization

Description

This feature allows users to manage their personal information, interests, and preferences.

Priority: High

Functional Requirements

FR-10: The system shall allow users to view their profile.

FR-11: The system shall allow users to update profile information.

FR-12: The system shall allow users to define reading interests.

FR-13: The system shall allow users to manage preferred subjects and categories.

FR-14: The system shall use user preferences to improve personalized recommendations.

4.3 Smart Library Search

Description

The system shall provide both traditional keyword search and AI-powered semantic search.

Priority: High

Stimulus/Response

1. User enters a query.
2. System determines the search mode.
3. Keyword search retrieves metadata matches.
4. Semantic search converts the query into an embedding.
5. Relevant resources are retrieved.
6. Results are ranked.
7. Results are displayed.

Functional Requirements

FR-15: The system shall support keyword-based resource search.

FR-16: The system shall support natural-language search.

FR-17: The system shall support AI-powered semantic search.

FR-18: The system shall rank search results based on relevance.

FR-19: The system shall allow filtering by title.

FR-20: The system shall allow filtering by author.

FR-21: The system shall allow filtering by category.

FR-22: The system shall allow filtering by subject.

FR-23: The system shall allow filtering by publication year.

FR-24: The system shall allow filtering by resource type.

FR-25: The system shall allow filtering by language and file format where metadata is available.

4.4 Digital Library and Resource Management

Description

This feature allows users to discover and access digital resources.

Priority: High

Functional Requirements

FR-26: The system shall display detailed resource information.

FR-27: Resource details shall include title, author, description, category, publication information, and resource type where available.

FR-28: The system shall allow authorized users to access supported digital resources.

FR-29: The system shall provide an integrated document viewer for supported documents.

FR-30: The system shall allow users to continue reading from their last recorded position.

FR-31: The system shall validate digital resource formats before processing.

4.5 Reading History and Favorites

Description

The system shall personalize user interaction through reading history and favorites.

Priority: Medium

Functional Requirements

FR-32: The system shall record resources accessed by users.

FR-33: The system shall maintain users' reading history.

FR-34: The system shall display recently accessed resources.

FR-35: The system shall record reading progress where applicable.

FR-36: The system shall allow users to bookmark resources.

FR-37: The system shall allow users to view and manage their favorites.

4.6 AI Chat Assistant

Description

The AI Assistant provides natural-language interaction with the library platform.

Priority: High

Functional Requirements

FR-38: The system shall provide an AI-powered conversational assistant.

FR-39: The AI Assistant shall support natural-language questions.

FR-40: The AI Assistant shall answer library-related queries.

FR-41: The AI Assistant shall provide contextual responses when sufficient information is available.

FR-42: The system shall indicate when sufficient information is unavailable.

4.7 RAG-Based Document Question Answering

Description

This feature allows users to ask questions about digital resources using Retrieval-Augmented Generation.

Priority: High

RAG Pipeline

```
Document Upload
      ↓
Text Extraction
      ↓
Document Chunking
      ↓
Embedding Generation
      ↓
ChromaDB
      ↓
User Question
      ↓
Query Embedding
      ↓
Relevant Chunk Retrieval
      ↓
LLM
      ↓
Grounded Answer
      ↓
Citation
```

Functional Requirements

FR-43: The system shall allow authorized users to query supported digital documents.

FR-44: The system shall extract textual content from supported documents.

FR-45: The system shall divide documents into searchable chunks.

FR-46: The system shall generate embeddings for processed document chunks.

FR-47: The system shall store document embeddings in the vector database.

FR-48: The system shall retrieve relevant document content for user queries.

FR-49: The system shall generate answers using retrieved content.

FR-50: The system shall provide document-based citations whenever applicable.

FR-51: The system shall indicate when relevant evidence cannot be found.

4.8 AI Document Summarization

Description

The system shall generate concise summaries of digital resources.

Priority: Medium

Functional Requirements

FR-52: The system shall generate summaries of supported books and documents.

FR-53: The system shall support summarization of research papers and technical documents.

FR-54: The system shall present summaries in a readable format.

FR-55: The system shall avoid presenting unsupported information as document content.

4.9 Personalized Recommendations

Description

The recommendation engine provides resources based on user interests and behavior.

Priority: High

Functional Requirements

FR-56: The system shall recommend resources based on user interests.

FR-57: The system shall use reading history where available.

FR-58: The system shall use favorite resources where appropriate.

FR-59: The system shall recommend resources based on content similarity.

FR-60: The system shall update recommendations as user preferences and behavior change.

4.10 Citation and Evidence Support

Description

The system shall provide evidence supporting AI-generated document answers.

Priority: High

Functional Requirements

FR-61: The system shall display the source document for document-based AI responses.

FR-62: The system shall display page numbers or relevant source locations when available.

FR-63: The system shall distinguish retrieved evidence from generated explanation.

4.11 Digital Study Hall

Description

Digital Study Hall provides collaborative learning and knowledge-sharing functionality.

Priority: High

Functional Requirements

FR-64: The system shall allow users to create Study Hall groups.

FR-65: The system shall allow users to join existing groups.

FR-66: The system shall support real-time group discussions.

FR-67: The system shall allow members to share resources.

FR-68: The system shall allow members to share notes and learning materials.

FR-69: The system shall display Study Hall activity.

FR-70: The system shall allow authorized moderators to remove inappropriate content.

4.12 Notifications

Description

The notification service keeps users informed about relevant system activity.

Priority: Medium

Functional Requirements

FR-71: The system shall notify users about relevant Study Hall activities.

FR-72: The system shall notify users about new resources where applicable.

FR-73: The system shall notify users about important system updates.

FR-74: The system shall allow users to view notifications.

FR-75: The system shall allow users to mark notifications as read.

4.13 Librarian Management

Description

The librarian manages digital library resources.

Priority: High

Functional Requirements

FR-76: The system shall allow librarians to add resources.

FR-77: The system shall allow librarians to edit resource information.

FR-78: The system shall allow librarians to remove outdated or invalid resources.

FR-79: The system shall allow librarians to upload PDFs, e-books, journals, and research papers.

FR-80: The system shall allow librarians to manage authors.

FR-81: The system shall allow librarians to manage categories.

FR-82: The system shall allow librarians to manage resource metadata.

FR-83: The system shall organize resources by subject, category, and resource type.

4.14 Content Verification and Moderation

Functional Requirements

FR-84: The system shall allow librarians to review user-uploaded resources where user uploads are enabled.

FR-85: The system shall allow librarians to approve or reject submitted resources.

FR-86: The system shall prevent unapproved user content from being publicly published.

FR-87: The system shall allow librarians to moderate Study Hall discussions.

FR-88: The system shall allow librarians to remove inappropriate content.

4.15 Reports and Analytics

Description

The analytics module supports data-driven library administration.

Priority: Medium

Functional Requirements

FR-89: The system shall generate resource usage reports.

FR-90: The system shall display popular resources.

FR-91: The system shall display active user statistics.

FR-92: The system shall display reading activity.

FR-93: The system shall display AI usage statistics.

FR-94: The system shall display Study Hall participation statistics.

FR-95: The system shall provide analytics dashboards for authorized administrators and librarians.

4.16 Administrator Management

Functional Requirements

FR-96: The system shall allow administrators to create user accounts where required.

FR-97: The system shall allow administrators to update user accounts.

FR-98: The system shall allow administrators to suspend user accounts.

FR-99: The system shall allow administrators to delete user accounts subject to applicable policies.

FR-100: The system shall allow administrators to manage librarian accounts.

FR-101: The system shall allow administrators to assign roles.

FR-102: The system shall allow administrators to manage permissions.

FR-103: The system shall allow administrators to configure system-wide settings.

4.17 Security and Audit Management

Functional Requirements

FR-104: The system shall maintain audit logs of privileged administrative activities.

FR-105: The system shall record security-relevant events.

FR-106: The system shall restrict access to administrative functions according to roles.

FR-107: The system shall allow administrators to monitor security-related activity.

4.18 Document Processing and Vector Indexing

Functional Requirements

FR-108: The system shall validate uploaded documents.

FR-109: The system shall extract text from supported documents.

FR-110: The system shall preprocess document content for indexing.

FR-111: The system shall generate vector embeddings for processed documents.

FR-112: The system shall store embeddings in the vector database.

FR-113: The system shall update the vector index when an approved document changes.

FR-114: The system shall remove corresponding vector data when a resource is permanently removed.

5. Other Nonfunctional Requirements

The earlier SRS used general quality statements, but the updated version makes them more measurable and aligned with the new architecture.

5.1 Performance Requirements

NFR-01: The system shall respond to normal non-AI user requests within approximately 2 seconds under normal operating conditions.

NFR-02: Semantic search shall return results within approximately 5 seconds under normal operating conditions.

NFR-03: Standard AI responses should be generated within approximately 8 seconds under normal conditions, excluding external service outages and unusually large documents.

NFR-04: The system shall support multiple concurrent users without significant performance degradation.

NFR-05: Search results shall be displayed progressively where possible to improve perceived responsiveness.

NFR-06: Document indexing shall operate asynchronously where appropriate so that large document processing does not block normal library operations.

5.2 Safety Requirements

Although AstraLib is not a safety-critical system, it shall protect the integrity of user and library information.

NFR-07: The system shall prevent accidental deletion of important records through authorization controls and confirmation mechanisms.

NFR-08: The system shall maintain regular backups.

NFR-09: The system shall prevent unauthorized modification of library resources.

NFR-10: Error messages shall not expose passwords, API keys, database credentials, or internal system information.

NFR-11: The system shall provide recovery procedures following system failure.

NFR-12: AI-generated information shall not be represented as authoritative when supporting evidence is unavailable.

5.3 Security Requirements

NFR-13: Passwords shall never be stored as plain text.

NFR-14: The system shall use secure password hashing.

NFR-15: The system shall implement JWT-based authentication or an equivalent secure mechanism.

NFR-16: The system shall implement role-based authorization.

NFR-17: All client-server communication shall use HTTPS in production.

NFR-18: The system shall validate and sanitize user input.

NFR-19: The system shall protect against common web security threats.

NFR-20: API credentials shall be stored securely on the server side.

NFR-21: The frontend shall not expose private AI API keys.

NFR-22: Sessions/tokens shall expire according to configured security policies.

NFR-23: Administrative actions shall be recorded in audit logs.

NFR-24: Uploaded files shall be validated before processing.

5.4 Software Quality Attributes

Reliability

NFR-25: The system shall maintain consistent operation during normal usage.

NFR-26: The system shall handle service failures gracefully.

NFR-27: Core library functions shall remain available when AI services are temporarily unavailable, where technically feasible.

Usability

NFR-28: The system shall provide an intuitive user interface.

NFR-29: Navigation shall be consistent across modules.

NFR-30: AI responses shall be presented in a clear and understandable format.

NFR-31: Important system errors shall be displayed using understandable messages.

Accessibility

NFR-32: The system should follow WCAG 2.1 Level AA accessibility principles where applicable.

NFR-33: Major functions shall support keyboard navigation.

NFR-34: Text, buttons, forms, and navigation elements shall use accessible labels and sufficient visual contrast.

Scalability

NFR-35: The system shall support growth in the number of users and digital resources.

NFR-36: The AI service shall be independently scalable from the core backend.

NFR-37: The architecture shall support multiple simultaneous Study Hall groups.

NFR-38: The AI layer shall allow integration with additional LLM providers in the future.

Maintainability

NFR-39: The system shall follow modular architecture.

NFR-40: Source code shall be documented.

NFR-41: Source code shall be version-controlled.

NFR-42: Individual modules shall be independently testable.

NFR-43: AI services shall be separated from core business logic to simplify maintenance.

Portability

NFR-44: The application shall operate on Windows, Linux, and macOS through supported browsers.

NFR-45: The application shall support mobile and tablet browsers through responsive design.

Availability

NFR-46: The production system should target 24×7 availability except during planned maintenance.

NFR-47: System health shall be monitored.

NFR-48: Critical failures shall be logged for troubleshooting.

AI Quality

NFR-49: RAG-generated document answers shall be based on retrieved document content.

NFR-50: The system shall provide source citations whenever applicable.

NFR-51: The system shall indicate when sufficient evidence is unavailable.

NFR-52: The system shall avoid intentionally generating unsupported document claims.

NFR-53: AI functionality shall degrade gracefully if the external AI service is unavailable.

NFR-54: AI-generated content shall clearly be identified as AI-generated where appropriate.

5.5 Business Rules

BR-01: Every user must authenticate before accessing protected services.

BR-02: Users can access only functions permitted by their assigned role.

BR-03: Only authorized librarians and administrators can modify library resource metadata.

BR-04: Only authorized librarians can approve user-submitted resources.

BR-05: Unapproved resources shall not be publicly discoverable when moderation is enabled.

BR-06: Administrators have the highest system privilege.

BR-07: Users can manage their own favorites and reading history.

BR-08: Users cannot modify another user's private information.

BR-09: AI document answers must use retrieved document evidence where the question is document-specific.

BR-10: The AI system shall indicate when relevant evidence cannot be found.

BR-11: Deleted resources shall no longer appear in normal search results.

BR-12: Corresponding vector representations shall be removed or invalidated when resources are permanently deleted.

BR-13: Study Hall content must follow the platform's usage and moderation policies.

BR-14: Administrative operations shall be auditable.

BR-15: Digital resources shall be stored and distributed only according to applicable access and copyright policies.

6. Other Requirements

6.1 Database Requirements

AstraLib shall use MySQL as its primary relational database. The database shall maintain information related to:

User Data

- User ID
- Name
- Email
- Password hash
- Role
- Profile information
- Interests

Library Data

- Book/resource ID
- Title
- Author
- Description
- Category
- Subject
- Publisher
- Publication year
- Language
- Resource type
- File metadata

User Activity

- Reading history
- Reading progress
- Favorites
- Search history where enabled

Study Hall

- Groups
- Members
- Discussions
- Shared resources
- Notes

Notifications

- Notification ID

- Recipient
- Message
- Timestamp
- Read status

Administration

- Roles
- Permissions
- Audit logs
- System settings

6.2 AI and RAG Requirements

AstraLib's AI layer shall provide intelligent knowledge discovery.

Semantic Search

The system shall: receive natural-language queries; generate query embeddings; search vector representations; retrieve semantically relevant resources; and rank and display results.

RAG

The system shall: process documents; extract text; split text into chunks; generate embeddings; store embeddings in ChromaDB; retrieve relevant chunks; provide retrieved content to the LLM; generate a grounded response; and provide source information where available.

AI Assistant

The AI Assistant shall support natural-language interaction with users.

Summarization

The system shall summarize supported documents.

Recommendation

The system shall use user interests, reading history, favorites, and content similarity to generate relevant recommendations.

6.3 File Storage Requirements

Digital files shall not be stored directly inside relational database records where unnecessary. File storage shall support:

- PDF
- E-books
- Research papers
- Journals
- Technical documents
- Notes
- Other approved formats

The system shall maintain metadata linking each stored file to its corresponding database record.

6.4 Backup and Recovery Requirements

OR-01: The system shall perform periodic backups of critical database information.

OR-02: Backup files shall be stored securely.

OR-03: The system shall provide restoration procedures.

OR-04: Backup operations shall not unnecessarily interrupt normal library services.

OR-05: Administrators shall have appropriate permissions to initiate or manage restoration.

6.5 Accessibility Requirements

AstraLib should support inclusive access for users from different backgrounds and disciplines. The system should provide:

- Responsive design
- Keyboard navigation
- Accessible forms
- Clear typography
- Appropriate contrast
- Screen-reader-friendly labels
- Simple navigation
- Clear AI responses

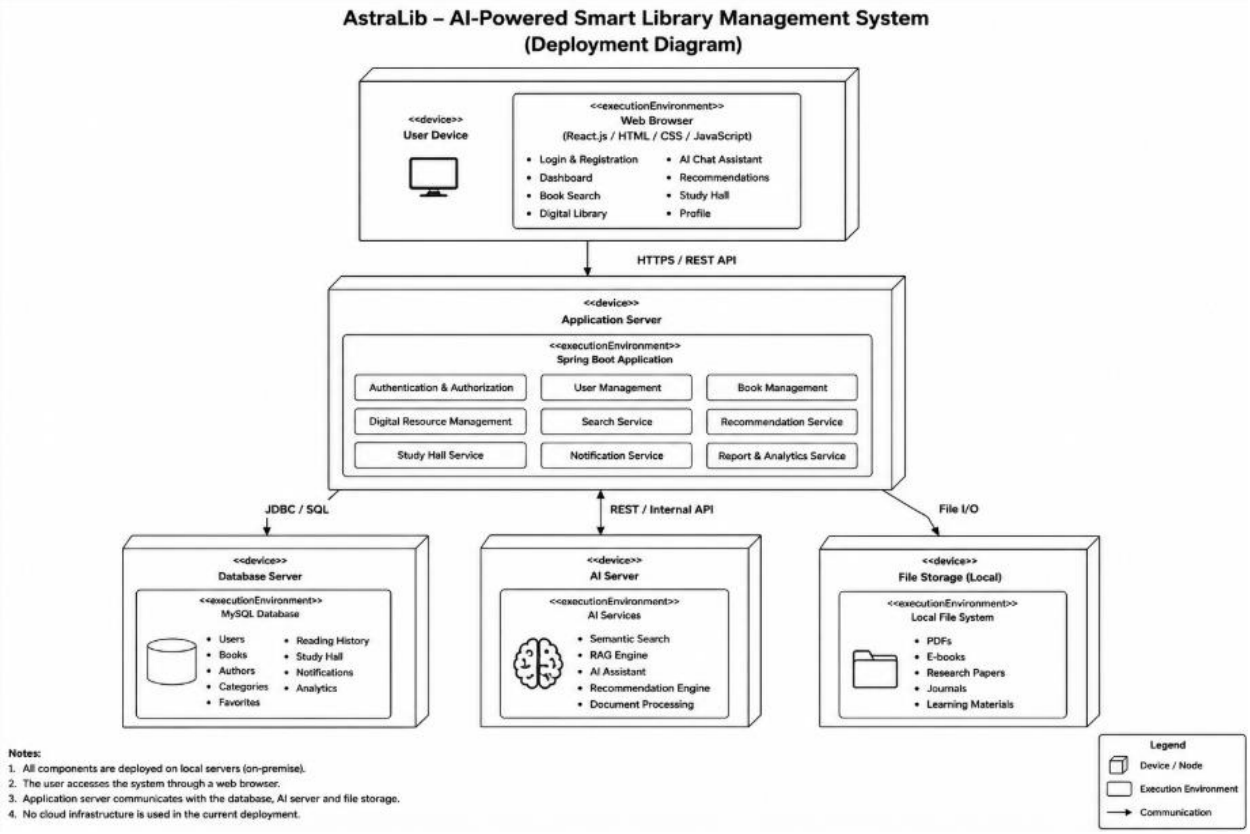
Multilingual interaction is considered a future enhancement unless included in the initial implementation scope.

6.6 Browser and Platform Requirements

The system shall support Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The application shall be responsive across desktop, laptop, tablet, and mobile.

6.7 Deployment Requirements

The production architecture shall follow a layered deployment model, as shown below.



The system may be deployed using cloud infrastructure. The final hosting provider can be selected during implementation according to cost, free-tier availability, performance, storage requirements, database support, and deployment compatibility.

6.8 Future Enhancements

The following features may be incorporated in future versions.

Knowledge Graph

A knowledge graph can represent relationships among books, authors, topics, subjects, concepts, and research areas.

Multilingual Interaction

Future versions may support multiple interface languages, multilingual search, and multilingual AI responses.

Voice Assistant

Users may be able to search by voice, ask AI questions by voice, and receive spoken responses.

OCR

OCR can allow scanned documents to become searchable.

Mobile Application

Native Android and iOS applications may be developed.

Advanced Recommendation

Future versions may use more sophisticated collaborative and hybrid recommendation models.

Advanced Analytics

Future analytics may include knowledge discovery trends, user behavior analysis, resource demand prediction, and topic trends.

External Knowledge Sources

Future versions may integrate additional scholarly and library APIs.

Appendix A: Glossary

Artificial Intelligence (AI)

Technology that enables software systems to perform tasks involving capabilities such as language understanding, reasoning, prediction, and generation.

AstraLib

The proposed AI-powered universal knowledge discovery and library management platform.

API

Application Programming Interface used for communication between software components.

Backend

The server-side part of the application responsible for business logic, data management, security, and APIs.

ChromaDB

A vector database used to store and retrieve vector embeddings for semantic retrieval.

Digital Resource

A digital information resource such as an e-book, PDF, journal, research paper, technical document, or note.

Embedding

A numerical vector representation of text or other data that captures semantic relationships.

FastAPI

A Python framework used to build APIs, including AI services.

LLM

Large Language Model used to understand and generate natural-language responses.

RAG

Retrieval-Augmented Generation, an architecture in which relevant external information is retrieved before generating an AI response.

Semantic Search

A search technique that retrieves information based on meaning and context rather than only exact keyword matching.

Vector Database

A database designed to store and search high-dimensional vector representations.

Librarian

A privileged user responsible for managing digital library resources and moderating content.

Administrator

A user with the highest level of system privileges.

General User

Any authorized person using AstraLib to discover, read, research, or learn from digital resources.

Study Hall

A collaborative digital environment where users with shared interests can communicate and learn together.

JWT

JSON Web Token used for secure authentication and authorization.

RBAC

Role-Based Access Control, a mechanism that restricts access based on user roles.

Appendix B: Analysis Models

The following analysis and design models shall be developed as part of the AstraLib Software Design Document:

- System Model
- System Architecture Diagram
- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Entity-Relationship Diagram
- Class Diagram
- Data Flow Diagram – Level 0
- Data Flow Diagram – Level 1
- Data Flow Diagram – Level 2
- State Diagram
- Deployment Diagram
- Component Diagram
- Module Diagram

The diagrams shall be derived directly from the requirements defined in this SRS.

Appendix C: To Be Determined List

The following items may be finalized during implementation:

TBD-01: Final LLM provider and model version.

TBD-02: Final embedding model.

TBD-03: Final cloud hosting provider.

TBD-04: Final cloud file-storage provider.

TBD-05: Exact real-time communication technology for Study Hall.

TBD-06: Final recommendation algorithm.

TBD-07: Exact supported e-book formats beyond PDF.

TBD-08: Email verification provider.

TBD-09: Exact backup frequency and retention policy.

TBD-10: Final multilingual models and supported languages.

TBD-11: Knowledge graph implementation technology.

TBD-12: OCR technology for scanned documents.

Appendix D: Final Requirement Traceability

Project Area	Corresponding SRS
User Management	FR-01–14
Smart Search	FR-15–25
Digital Library	FR-26–31
History & Favorites	FR-32–37
AI Assistant	FR-38–42
RAG	FR-43–51
Summarization	FR-52–55
Recommendations	FR-56–60
Citations	FR-61–63
Study Hall	FR-64–70
Notifications	FR-71–75
Librarian	FR-76–83
Moderation	FR-84–88
Analytics	FR-89–95
Administration	FR-96–103
Security/Audit	FR-104–107
Document Processing	FR-108–114
Performance	NFR-01–06
Safety	NFR-07–12
Security	NFR-13–24
Quality	NFR-25–48
AI Quality	NFR-49–54